

Daily Problem

November 25, 2025

Find:

$$\int \frac{1+x}{x(1+xe^x)} dx.$$

Solution:

$$\int \frac{1+x}{x(1+xe^x)} dx = \int \frac{e^x(1+x)}{e^x x(1+xe^x)} dx$$

notice that:

$$(e^x x)' = e^x(1+x)$$

so,

$$\int \frac{e^x(1+x)}{e^x x(1+xe^x)} dx = \int \frac{1}{e^x x(1+xe^x)} de^x x = \ln |xe^x| - \ln |1+xe^x| + C$$